



Nonlinear aeroelastic scaling of high aspect-ratio wings



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ABSTRACT

The aeronautical industry is currently facing the simultaneous and conflicting demand to enhance flight efficiency while reducing emissions. One potential solution for reducing fuel consumption is to increase the wing aspect-ratio as it improves the lift-to-drag ratio. However, higher aspect-ratio wings result in higher deflections which in turn may lead to nonlinear aeroelastic behavior. In this work, the aeroelastic behavior of a conventional regional aircraft with high aspect-ratio wings is investigated. Aeroelastically scaled models using different scaling methodologies have been evaluated and compared. These methodologies use scaling factors derived from the governing aeroelastic equations of motion to set the target values to be matched through the optimization of the scaled model structure. Two linear scaling approaches were used: the first method consists of a direct modal response matching; while the second method uncouples the mass and stiffness distribution to achieve the modal response. An alternative nonlinear aeroelastic scaling methodology using equivalent static loads is presented, which uses two different optimization routines to match the nonlinear static response and the mode shapes of the full model. The aeroelastic response agreement was found to be considerably better when the nonlinear approach is applied and the accuracy is noticeably better than the results obtained using the traditional linear scaling methods.

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1. Introduction

Presently, the original aircraft manufacturers (OEM) are looking at new aircraft designs with high aspect-ratio wings, especially for civil and commercial aircraft. High Aspect-Ratio Wings (HARW) present considerable performance advantages. In general, HARW produce more lift and provide a higher lift-to-drag ratio resulting in increased endurance [1]. Also, HARW increase not only aircraft stability but also efficiency because they produce less induced drag leading to lower fuel consumption. To make HARW feasible in terms of weight penalties, these wings are designed to be very flexible.

The total drag coefficient can be defined [1] as

$$C_D = C_{D_0} + C_{D_i} + C_{D_w} = C_{D_0} + \frac{C_L^2}{\pi e AR} + C_{D_w}, \quad (1)$$

where C_D , C_{D_0} , C_{D_i} , C_{D_w} are the coefficients of total drag, profile drag, induced drag and wave drag, respectively; C_L is the lift coefficient; e the span efficiency factor and AR the wing aspect-ratio.

To make HARW feasible in terms of weight penalties, these wings are designed to be very flexible. Nevertheless, this challenge warrants further investigation due to large deformations under normal operating loads leading to a geometrical nonlinear behavior and aeroelastic problems [2,3]. These large deformations can change the natural frequencies of the wing which can produce noticeable changes in its aeroelastic behavior [4].

In order to better understand the physical behavior of the wing without building an expensive full scale demonstrator, a reduced scale model provides a feasible alternative. Experimental testing of aeroelastically scaled models is a common approach in new flight vehicle development programs [5]. In this work, the scaled model is intended to closely reproduce the aeroelastic response of the full scale model at operating conditions.

Aeroelastic scaling requires adequate consideration to aerodynamic and structural physics. Aerodynamic similitude is achieved analytically by geometrically scaling the aerodynamic shape when Mach and Reynolds number are consistent. Flight conditions such as airspeed and altitude are selected for matching scaled parameters like Froude number and density ratio, for example. The Froude number is a dimensionless parameter defined as the ratio between inertial and gravitational forces [6]:

$$Fr = \frac{V}{\sqrt{ba_g}}, \quad (2)$$

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Nomenclature

AR	Wing aspect-ratio	n	Load factor
a_g	Acceleration of gravity	n_F	Force ratio
b	Reference wing span	n_f	Frequency ratio
C_D	Total drag coefficient	n_g	Length ratio
C_{D_0}	Profile drag coefficient	n_m	Mass ratio
C_{D_i}	Induced drag coefficient	n_v	Velocity ratio
C_{D_w}	Wave drag coefficient	n_ρ	Air density ratio
C_L	Lift coefficient	L	Total lift
e	Span efficiency factor	$[Q]$	Aerodynamic coefficient matrix
F	Force	S	Wing area
Fr	Froude number	u	Static deflection
f	Natural frequency	V	Air speed
g	Damping factor	W	Aircraft weight
$[K]$	Stiffness matrix	x	System degrees of freedom
k	Reduced frequency	α	Angle of attack
M	Mass	ρ	Air density
$[M]$	Mass matrix	ϕ	Natural mode shape

where b is the reference wing span, V the airspeed and a_g the gravity acceleration. The density ratio, n_ρ , is the ratio between the air density at which the scaled model will operates (ρ_s) and the air density at which the full model operates (ρ_f) [7]:

$$n_\rho = \frac{\rho_s}{\rho_f} . \quad (3)$$

Structural similitude is not realistically achievable by geometrically scaling the structural components and corresponding analytical scaling requirements will generally specify that a geometrically scaled structure should be made from materials that have non-physical properties. There is also a high probability that the manufacturing techniques used for the full scale design cannot be duplicated at a smaller scale. The only feasible option is to redesign the internal structure using available materials and optimize it such that its scaled mass and stiffness properties are consistent with the full scale aircraft [8]. The ladder structure is one of several scaled model configurations recommended by Bisplinghoff et al. [7].

In the classical approach, aeroelastic scaling is achieved by selecting a discrete subset of modal degrees of freedom that capture the relevant global properties of the full model, and optimizing the scaled aircraft such that the non-dimensional modal masses and stiffness coefficients match the full scale aircraft [7].

The most common practice for classical aeroelastic scaling is to use a truncated number of the vibration mode shapes from the target full scale model as the modal degrees of freedom for the scaled model optimization [9]. However, there is a drawback. The truncation may omit information that becomes important when geometric nonlinearities are significant (it can be considered analogous to omitting certain flexibility in the model (e.g., axial and shear)). Classical scaling methods have worked in practice for traditional applications, but the validity of the modeling assumptions needs verification for cases where geometric nonlinearities become important.

Few papers can be found in the literature regarding nonlinear aeroelastic scaling [10–13]. Just recently this topic has become a focus of study in order to investigate the High Altitude Long Endurance (HALE) aircraft, which are characterized by their high aspect-ratio wings and high structural flexibility, either by using a Joined-Wing configuration [10,11,13] or a conventional configuration [12].

Bond et al. [11] developed a nonlinear aeroelastic scaling methodology similar to the one presented by French and Eastep [5]. This methodology directly matches the first three natural fre-

quencies and corresponding mode shapes, and the first buckling eigenvalue. From the results retrieved, the authors reported a good matching for the aeroelastic frequencies and damping.

Wan and Cesnik [12] devised scaling parameters to incorporate geometric nonlinearities (through nonlinear stiffness matrix) and pre-stress to be applied to unsteady aeroelastic system. The main observations drawn by the authors were the following: similar scaling laws are also applicable to scale down structures presenting geometric nonlinearities; the Froude number should not be disregarded when considering high aspect-ratio wings; matching the Reynolds number (when Reynolds number is low) may result in an additional challenge to other scaling factors.

Simultaneously, Ricciardi et al. [13] proposed a new aeroelastic scaling methodology capable of incorporating geometric nonlinearities in the scaling process. With optimization efficiency in mind Equivalent Static Loads (ESL) were included in their methodology. From the comparison of their scaling methodology with a classic scaling methodology, the authors verified that the deflections and aeroelastic frequencies results were improved, although with a loss of accuracy in terms of flutter speed.

In this work, a new nonlinear scaling methodology similar to the one proposed by Ricciardi et al. [13] is presented. The main difference is in what concerns to the way the scaling is performed: the stiffness and mass distributions are achieved separately in two optimization loops in the new approach; while in Ricciardi et al. method just one optimization procedure is carried out. The application case is a high aspect-ratio regional transport aircraft wing, which has a much lower aspect-ratio than those of the HALE aircraft of the previously reported nonlinear aeroelastic scaling works. Thus, this wing has a lower flexibility and smaller deformations than the aforementioned HALE aircraft. Despite this fact, geometric nonlinearities effects are clearly visible even for a 1 g load. Important considerations such as engine mass and pylon, besides fuel (no sloshing effect was modeled in the aeroelastic analyses) were taken into account in this work.

2. Theoretical background

Due to the complex nature of the problem, a simplified physics model was chosen: the small disturbance, linear potential partial differential equations (PDE) [14] as stated in Eq. (4):

$$[M]\{\ddot{x}\} + [K]\{x\} = \frac{\rho V^2}{2} [Q]\{x\} , \quad (4)$$

Table 1

Scaling factors, the subscripts of s and f represent scaled and full scale, respectively.

Dimension	Scaling factor
Length	$n_g = \frac{b_s}{b_f}$
Air density	$n_\rho = \frac{\rho_s}{\rho_f}$
Velocity	$n_v = \frac{V_s}{V_f}$
Mass	$n_m = \frac{M_s}{M_f} = n_g^3 n_\rho$
Frequency	$n_f = \frac{f_s}{f_f} = \frac{n_v}{n_g}$
Force	$n_F = \frac{F_s}{F_f} = n_\rho n_v^2 n_g^2$

where $[M]$, $[K]$, $[Q]$ are the mass, stiffness and aerodynamic coefficient matrices of the system, respectively; x are the degree of freedom of the system and V the airspeed. Although the physics is simplified, the PDE equations are adequate for a low-cost exploration of flight mechanics in real world conditions [14].

2.1. Scaling factors

The geometrically scaled model is constrained by a set of scaling parameters. These scaling factors are derived from non-dimensional parameters, which appear in the equations that govern the aerodynamics, the structure and their coupling (Eq. (4)). The choice of these scaling parameters is subject to constraints posed by the wind tunnel or environment in which the model will operate.

For the purpose of obtaining a non-dimensional set of equations, the designer has to select three primary quantities, which are the only parameters that can be changed. There is a wide range of choices that can be made, however only one combination is investigated here:

- the span b (based on the desired reduced scale span, it is due to wind tunnel constraints);
- air density ρ (it is due to environmental constraints);
- velocity V (it is due to wind tunnel constraints).

These are the primary quantities used in the Buckingham's Π theorem [15] that is applied to the other important properties, which are needed for good aeroelastic similitude between full scale and reduced scale models. The factors used to attain to the aeroelastic scaling are summarized in Table 1.

The frequency is generally obtained using the reduced frequency parameter, k , which is defined as [9]:

$$k = \frac{f b}{V}, \quad (5)$$

where f is the natural frequency, b the wing reference span and V the airspeed. The frequency ratio then becomes:

$$\left(\frac{f b}{V}\right)_s = \left(\frac{f b}{V}\right)_f \Rightarrow n_f = \frac{f_s}{f_f} = \frac{V_s}{V_f} \frac{b_f}{b_s} = \frac{n_v}{n_g}. \quad (6)$$

The velocity is a “free-parameter”, however it must be constrained by the requirement that both full scale and scaled models operate at the same trim state, i.e., the same C_L . From the scaled factors summarized in Table 1, the lift ratio L_s/L_f can be scaled either by the force ratio (n_F) or the mass ratio (n_M) as demonstrated on Eqs. (7) and (8):

$$\frac{L_s}{L_f} = \frac{\left(\frac{\rho V^2 S C_L}{2}\right)_s}{\left(\frac{\rho V^2 S C_L}{2}\right)_f} = n_\rho n_v^2 n_g^2, \quad (7)$$

$$\frac{L_s}{L_f} = \frac{W_s}{W_f} = n_M. \quad (8)$$

Therefore from the lift equation the velocity is evaluated:

$$n_M = n_\rho n_v^2 n_g^2 \Rightarrow n_v = \frac{V_s}{V_f} = \sqrt{\frac{n_M}{n_\rho n_g^2}}. \quad (9)$$

The Froude number is a ratio between inertial and gravitational forces, which in turn determines the ratio of deflections due to gravity compared to those due to aerodynamic forces. In many cases, the deflection due to gravity can be assumed negligible compared to the aerodynamic forces. If this assumption is valid, the Froude number can be ignored (this is often the case for wind tunnel models). However, in the case of the unconstrained model in free flight, the Froude number may play a more significant role [14]. Also for high aspect-ratio wings where the flexibility is relevant as verified by Wan and Cesnik [12], the Froude number is an important factor. In order to maintain a similitude of Froude number, the following ratio (see Eq. (10)) must be maintained:

$$\left(\frac{V}{\sqrt{b a g}}\right)_s = \left(\frac{V}{\sqrt{b a g}}\right)_f. \quad (10)$$

The task of matching the Reynolds number at a desired velocity-altitude is a difficult one. It requires that the viscosity of the surrounding environment be changed, which in most cases is not easily accomplished. For this reason, the matching of the Reynolds number will likely not be satisfied. Although this may seem of concern, experience has shown that it is generally of minor importance where aeroelastic effects for main lifting surfaces are concerned [16].

The Mach number is an important parameter if compressibility effects are a phenomenon intended to be captured by the scaled model. Like the Reynolds number, the Mach number can be difficult to achieve if velocity and air density are chosen as primary quantities. Often it is assumed to have a small influence [13].

The scaling based on simplified physic model permits a reduction of the model restrictions, which would be much more complex if a scaling based on the Navier–Stokes equations was chosen.

2.2. Scaling methodologies

One design approach for developing accurate mass and stiffness distribution is to scale down the exact geometry of the full scale model. Unfortunately, this is not practical for several reasons. For instance, the manufacturing of scaled components may be very expensive and in some cases impossible due to their resulting small values and high precision measurements.

There is a need to find an alternative to the exact scaling of the full scale model geometry in order to achieve a match between the two models in terms of mass and stiffness distributions while at the same time ensuring manufacturability [9]. From the governing equations of linear elasticity, it can be shown that every model, with the same scaled geometry and the same scaled mass and stiffness distributions, will result in the same modal response (mode shapes and frequencies) as those of the full model, after appropriate scaling [17]. The target values (mass, inertia moments, natural frequencies, mode shapes and aerodynamic shape) for the scaled models are obtained from the full scale model.

To achieve the target values, different dynamic scaling methodologies can be used. For convenience, the ones that consider the analysis as linear will be classified as Linear Approach, while the methodologies that account for geometrical nonlinearities will be labeled as Nonlinear Approach.

In order to attain dynamic similitude, two different linear scaling methodologies can be used. The first method matches the

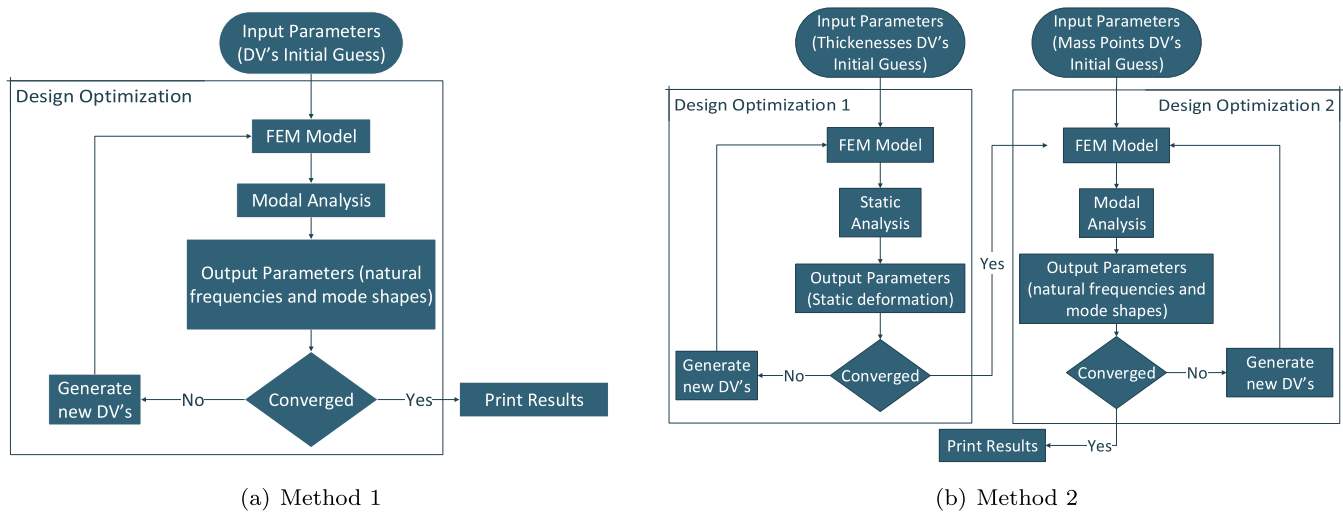


Fig. 1. Scaling methodologies.

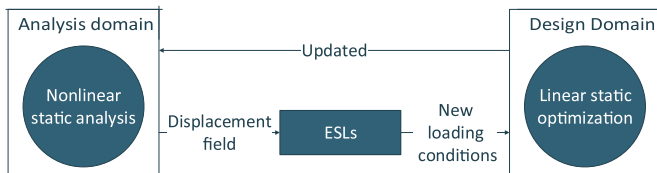


Fig. 2. Schematic process between the analysis and the design domain respectively (adapted from [18]).

modal response directly (Method 1) while the second method matches the modal response through a mass and stiffness distribution uncoupling (Method 2). Fig. 1 shows the flowcharts of both methodologies. The former methodology is a classical approach similar to the one presented by Bisplinghoff et al. [7], while the latter methodology was proposed by French and Eastep [5].

3. Nonlinear approach

Direct optimization of the nonlinear static response is a computationally intensive task. It is more efficient to use an equivalent linear system based on the Equivalent Static Loads (ESL) approach [18].

The ESL is defined as the static load that generates the same displacement field by a nonlinear static analysis which requires an iterative procedure that updates the linearized system as the design changes. The optimization process is divided into two domains: analysis and design, respectively. Non-linear static analysis is performed in the analysis domain. The nonlinear displacement field is evaluated and the equivalent load sets are computed in order to ensure the same displacement field between nonlinear and linear static analysis. The equivalent load sets are transferred to the design domain. In the design domain, linear static optimization is performed using the equivalent loads as external loads. The design variables are updated in the design domain and nonlinear static analysis is performed again with the update design variables. The process proceeds until the convergence criterion is satisfied. Note that this process is a multi-fidelity approach since the analyses conducted in analysis and design domains are of high fidelity and low fidelity, respectively. The basic idea of ESL is presented in Fig. 2, and a more detailed explanation of this method can be found in the literature [18].

The scaling methodology proposed for this work (Method 3) that includes geometric nonlinearities is shown in Fig. 3. In this new approach, the aeroelastic response of the scaled model will be attained by matching stiffness and mass distributions separately.

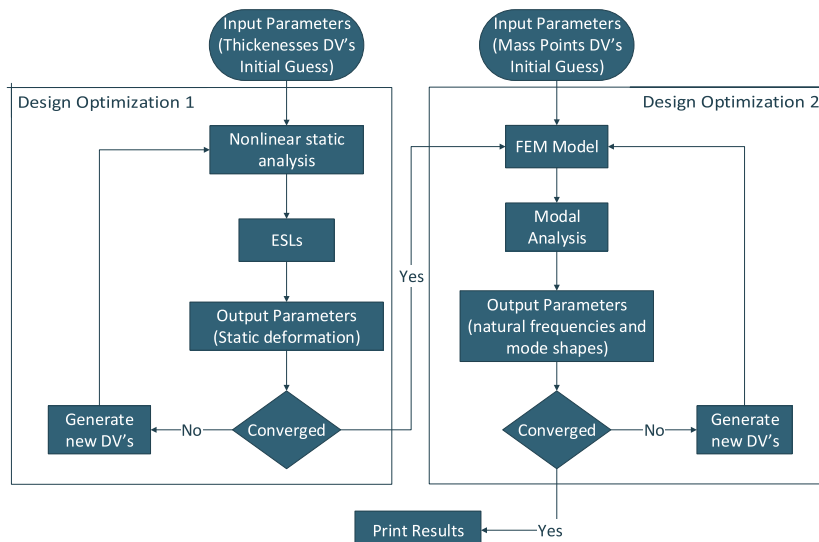


Fig. 3. Scaling methodology including geometric nonlinearities (Method 3).

The differences between this new methodology (Method 3, see Fig. 3) and Method 2 (see Fig. 1 (b)) rests essentially in the evaluation of the first routine, and it is also similar to the one proposed by Ricciardi et al. [13]. A nonlinear static analysis of the scaled model is performed in order to evaluate the nonlinear displacement field. The ESL approach is used to perform a linear static optimization of the scaled model. The optimization process in the Design domain (see Fig. 2) is performed in MatlabTM using the Sequential Quadratic Programming (SQP) algorithm. The static deflection error is calculated. This is the unique state variable of the design optimization which is applied and also the only parameter that takes part of the objective function to be minimized in the first routine; it can be stated as follows:

$$\text{minimize: } \sqrt{\sum_{i=1}^N \sum_{j=1}^3 \left(\left(\frac{t u_{i,j}^T - t u_{i,j}}{t u_{i,j}^T} \right)^2 + \left(\frac{r u_{i,j}^T - r u_{i,j}}{r u_{i,j}^T} \right)^2 \right)}$$

with respect to: structural design variables (e.g. spars and ribs thicknesses)

subject to: structural constraint (e.g. maximum stress lower than the maximum yield stress) (11)

where u^T and u are respectively the target and scaled model static deflections, the superscripts t and r stand for the translational and rotational parts of the deformed state, and N is the number of structural nodes used to represent the wing deformed shape.

When the structure stiffness distribution is achieved, concentrated masses are added to attain the desired mass distribution. The modal analysis is then performed, from which the modal results are obtained and the respective errors are evaluated. The errors are the state variables of the design optimization process and also take part of the objective function to be minimized in the second routine, as follows:

minimize:

$$\sqrt{\sum_{i=1}^M \left(\left(\frac{f_i^T - f_i}{f_i^T} \right)^2 + \sum_{j=1}^N \sum_{k=1}^3 \left(\left(\frac{t \phi_{i,j,k}^T - t \phi_{i,j,k}}{t \phi_{i,j,k}^T} \right)^2 + \left(\frac{r \phi_{i,j,k}^T - r \phi_{i,j,k}}{r \phi_{i,j,k}^T} \right)^2 \right) \right)}$$

with respect to: non-structural masses (e.g. mass points)

subject to: $\frac{\text{mass}^T - \text{mass}}{\text{mass}^T} - 0.05 \leq 0$

$$\begin{aligned} \frac{f_i^T - f_i}{f_i^T} - 0.05 &\leq 0 & \text{with } i = 1, \dots, M \\ \frac{t \phi_{i,j,k}^T - t \phi_{i,j,k}}{t \phi_{i,j,k}^T} - 0.05 &\leq 0 & \text{with } i = 1, \dots, M; \\ & & j = 1, \dots, N; \quad k = 1, \dots, 3 \\ \frac{r \phi_{i,j,k}^T - r \phi_{i,j,k}}{r \phi_{i,j,k}^T} - 0.05 &\leq 0 & \text{with } i = 1, \dots, M; \\ & & j = 1, \dots, N; \quad k = 1, \dots, 3 \end{aligned} \quad (12)$$

where natural frequencies and mode shapes are represented by the symbols f and ϕ , and M is the number of mode shapes selected for the scaling process. Once the objective modal result errors and convergence are achieved the final results are saved.

The main difference between the proposed methodology and the Ricciardi et al. [13] nonlinear scaling methodology is that in the former the stiffness and mass distribution matching is carried out in two separate optimization procedures, as in Method 2, while in the latter it is done in just one optimization loop, where a

Table 2

Main geometric characteristics of the modified NOVEMOR reference wing.

Parameter	Value
Span	36.33 m
Mean aerodynamic chord	3.02 m
First-half span (Wing 1)	7.13 m
Second-half span (Wing 2)	11.03 m
Root chord	5.71 m
Break chord	3.01 m
Tip chord	1.30 m

modal analysis and a prescribed number of aeroelastic static analyses are conducted.

4. Test case

For this work, the reference wing [19] of the EU 7th framework program project NOVEMOR [20] was used.

4.1. Full model

The full wing model is based on this reference wing, which was modified to exhibit a higher aspect-ratio behavior. The new wing was designed with an aspect-ratio of twelve while maintaining the wing area, the sweep and dihedral angle, the maximum take-off weight and the engine weight constant. The same cruise point of the unmodified wing geometry defined in the NOVEMOR project was selected: a Mach number of 0.78, and an altitude of 38,000 ft. The resulting main geometric characteristic of the wing are presented in Table 2.

Fig. 4 shows the geometrical and internal structural wing models used for calculations. The software used was MSC NastranTM.

On the baseline model, the wing contains a standard two-spar configuration placed at 25% and 75% chord. In order to maintain the internal configuration of the wing simple and to avoid further problems on the scaling of the full model, the pylon and the wing are modeled using equally spaced ribs. In the wing section, where the engine is placed, another rib was added with the purpose of increasing the wing stiffness. The wing spar/rib configuration is shown in Fig. 4(b). For the purpose of designing the wingbox, beam and shell elements are employed to model the wingbox caps and webs, respectively. Skins and ribs are modeled by shell elements.

The structural layout is designed to ensure structural integrity for an ultimate load of 3.8 g's. Linear and nonlinear static analyses are performed in order to evaluate the differences between the two analyses. For all the components of the structural model the material selected is the 7075 Aluminum alloy. Important differences between linear and nonlinear results are observed. Fig. 5(a) shows the differences in terms of the deflection at the wing tip when linear and nonlinear static analyses are performed for different values of load factor. The different load factors are achieved by increasing the aerodynamic load, by definition:

$$n = \frac{L}{W}, \quad (13)$$

where n is the load factor, L and W are the total lift and the total aircraft weight, respectively. The results diverge gradually when increasing the load factor. For an aerodynamic load causing a load factor 4 times higher than the gravity acceleration, the deflection at the wing tip is 2.4 times lower than what is predicted by linear analysis. Fig. 5 (b) and (c) shows the variation of the bending and twist angle respectively along the wing span for a load factor equal to one when the two analyses are performed. There is also a shortening effect on the wings which increases with the load factor increase, although this effect was found to be small enough

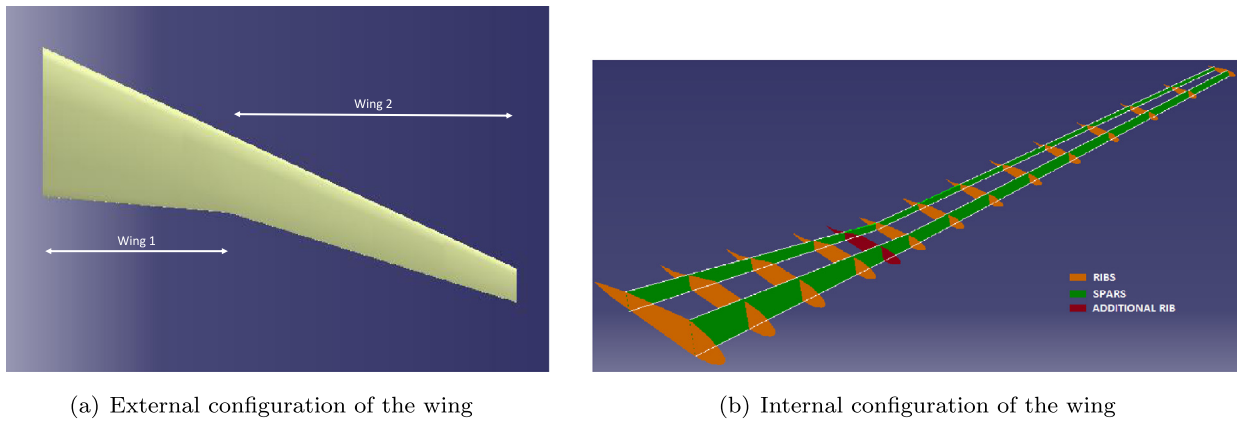
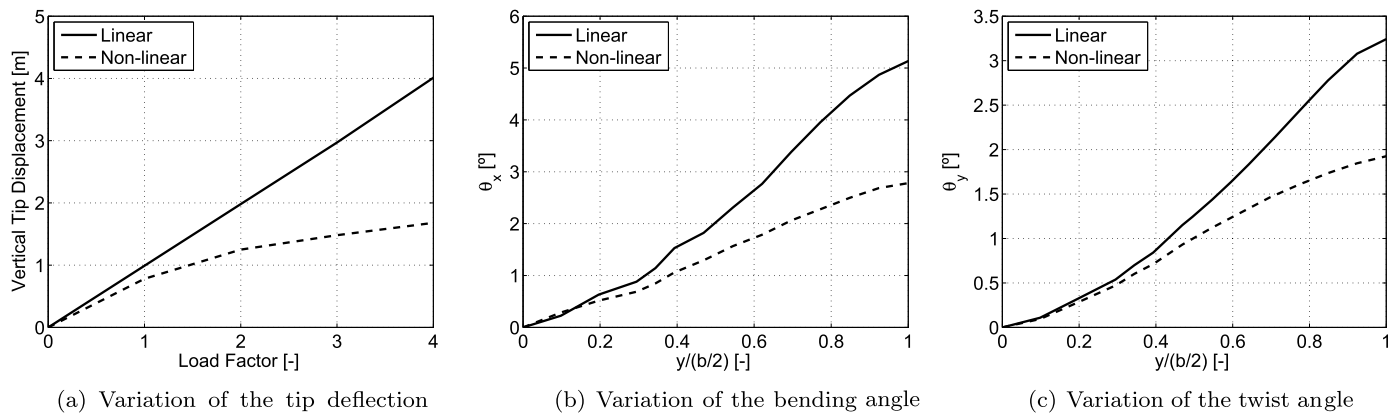


Fig. 4. Structural model.

Fig. 5. Variation of the tip deflection (a), bending (b) and twist angle (c) along the wing span ($n = 1$, where n is the load factor) in condition of empty tanks when linear and nonlinear analyses are performed.

(less than 1% of the wing semi-span) to be neglected. The results confirm the importance of accounting for geometric nonlinearities in the analysis of HARWs.

Both in Fig. 5(b) and in Fig. 5(c) it is observable a discontinuity in the break station ($y = 0.43 \cdot b/2$), which is due to a change in stiffness of the wing. From the root to the break station the wing is stiffer in order to support the engine.

Structural geometric nonlinearities can be attributed to the large displacements and the stress state (it is worth mentioning that there are other types of structural nonlinearities such as the contact and the buckling problems). The large displacements when coupled with a nonlinear solver that employs follower forces can substantially change the applied load and the stiffness matrix of the system and consequently changing the deformed state. The difference between the linear and nonlinear load and the linear and nonlinear stiffness matrix from which the last solution is computed increases as the structures suffers higher deformations, as it is observed in Fig. 5. Regarding the stress state, this effect has more influence when the material enters the plastic region, which can lead to a higher stiffness matrix and therefore lower displacements. However, no material nonlinearity was considered in the work.

4.2. Scaled model

Different scaled configurations were considered in the early stage of the scaling process, although in the end a configuration with a wing scaled as a ladder structure was selected (see Fig. 6). Rectangular spars placed at 25% and 75% chord and cylindrical rungs can be tailored to match the required stiffness. Beam ele-

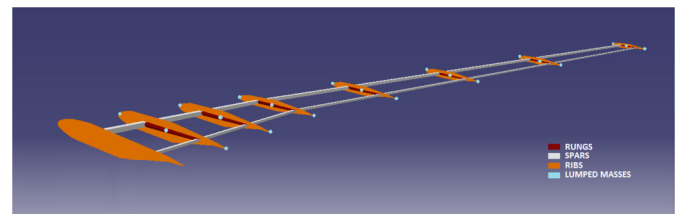


Fig. 6. Wing internal scaled structure.

ments adequately model the ladder structure. In order to maintain the aerodynamic shape of the airfoil, it was decided to model the wing skin with shell elements to simulate the behavior of a very thin layer (0.5 mm) of Balsa material. Balsa ribs would provide local support.

The pylon is an exact scale down of the full scale model and in accordance with the literature [21]; it is modeled by beam and shell elements in order to model the caps and webs, respectively.

In order to properly represent the aeroelastic behavior of the full model in the scaled model, it was decided to match the first seven natural frequencies and respective mode shapes (considering both translational and rotational parts) with a discrepancy of 5%. It was decided to use seven modes because the flutter phenomenon occurs as result of the coupling of the second flapwise bending¹ (fourth mode) and first torsion (seventh mode) natural modes of the wing. Also the overall mass should not exceed 5% of the error margin.

¹ Also known as flapwise or out of wing plane bending.

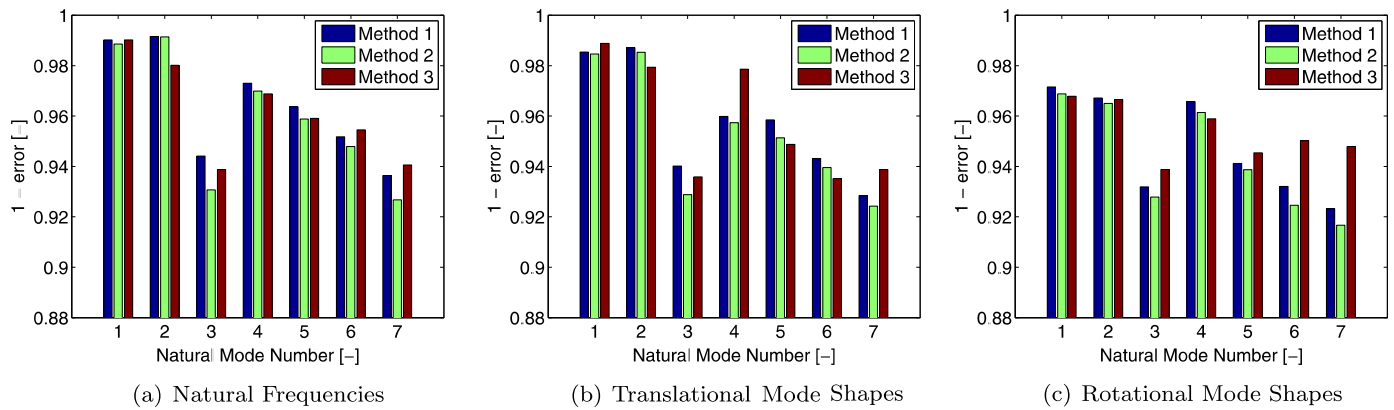


Fig. 7. Discrepancies between the three applied methods in order to match the first seven natural frequencies, translational and rotational mode shapes of the full model.

As far as the external geometry is concerned, the scaled model is an exact scale of the full scale model. A question is to find the best material to model the structural components of the scaled structure. An exact scale of the 7075 Aluminum alloy will lead to a non-existing isotropic material. Three materials (7075 Aluminum, Elektron 43 and Stainless Steel) were chosen for a comparison study, and from which the Elektron 43 was the one that presented the lowest average error in comparison with the target scaled material. Thus, it was selected to be the scaled model material and was followed in this work.

The design variables are the following parameters: width and height of the front and rear wing spars, diameter of wing rungs, size of the front and rear pylon spars, thickness of the ribs of the pylon and lumped masses. The constraints of the design are all the properties to be matched (mass, natural frequencies, mode shapes and in Methods 2 and 3 also the static deflection), which also take part of the objective function to minimize. Since the sum of the squares of the residuals (SSR) is a basic scheme for quantifying differences in the responses, it was used for the objective function.

5. Results

The performance results achieved by applying the three scaling methodologies are summarized in Table 3 (the detailed relative errors to the target values are shown in Table 4 below). The optimization did not succeed in achieving errors lower than 5% for all the defined target parameters, although the values reached were not far from it and the global average error for the three methodologies was also below 5%. The global average error consists in the average value of the errors of the target parameters, which include natural frequencies, mode shapes (translational and rotational parts) and mass. The optimization loops for all the applied methods were performed in MatlabTM using a Sequential Quadratic Programming (SQP) algorithm.

Starting with the comparison between both linear approaches (Method 1 and 2), it can be observed that Method 2 outperforms Method 1 in the computational effort, although with a small increase on the global average error (0.21%), namely on the modal behavior (natural frequencies and modes shapes) as can be seen in Fig. 7, where the first seven natural frequencies, translational and rotational mode shapes results achieved using the three different scaled methodologies are given in Fig. 7. The ordinate of the plots is $(1 - \text{error})$, where the error is the relative error of a frequency (or translational mode shape or rotational mode shape) value regarding the corresponding target frequency (or translational mode shape or rotational mode shape) value.

Method 1 is a more robust optimization process due to the coupling of mass and stiffness matrices, and with an advantage of using only one routine. Method 2 uses instead two routines,

Table 3

Results comparison between linear and nonlinear approaches. For methods 2 and 3, the computational time outside the brackets is the overall time, while the values between brackets correspond to the computational time of the first and second optimization routines.

Parameter	Method 1	Method 2	Method 3
Global average error	4.49%	4.70%	4.01%
Computational time*	29.1 h	17.4 h (7 h + 10.4 h)	55 h (47 h + 8 h)
Number of design variables	86	86	86

* The optimization process was performed with a personal computer with these characteristics: CPU = Intel(R) Core(TM)2 Duo CPU T6400 @ 2.00 GHz; RAM = 4.00 GB.

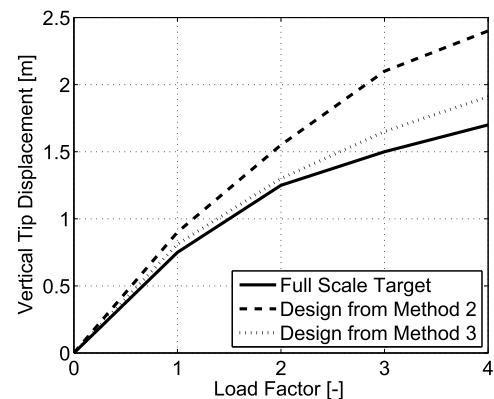


Fig. 8. Nonlinear tip deflection.

one for the stiffness distribution matching and the other for the mass distribution matching, but with better computational time efficiency. This is explained by the fact that by uncoupling the mass and the stiffness distributions simplified optimization routines are achieved, resulting in a decrease of the computational time required to attain good modal results.

The main purpose of applying Method 3 is to incorporate geometric nonlinearities in the scaling process, which is achieved with this methodology. When comparing the nonlinear static deformation for different prescribed loads, one observes that the scaled model designed with Method 3 matches the target more closely than the scaled model achieved using Method 2 (see Fig. 8). Tip wing deflection error is reduced from 14% to 11% at 4 g's case, for linear (Method 2) and nonlinear (Method 3) approach, respectively: the trend of the tip wing deflection is thus considerably closer to the target curve when the scaled model is obtained using the nonlinear approach.

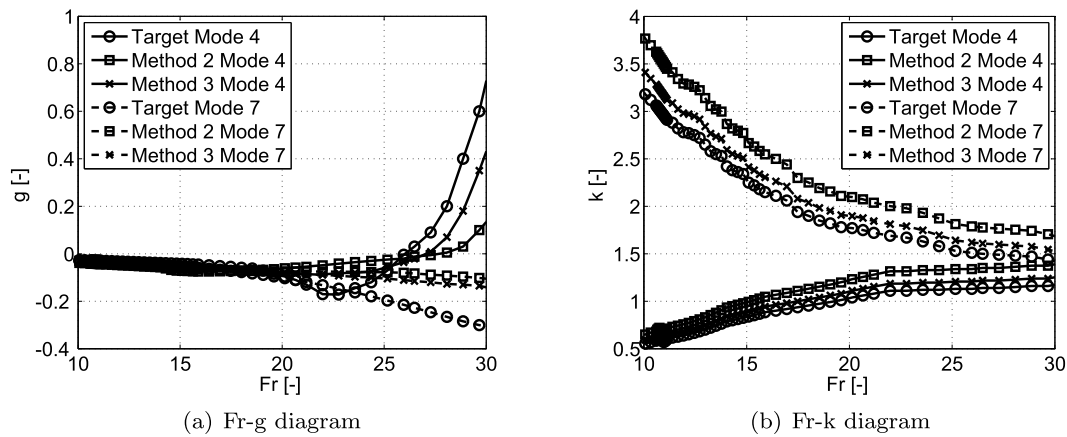


Fig. 9. Fr-g and Fr-k diagrams.

Table 4

Flutter speed and frequency differences between Method 2 and 3 in relation to the associated target scaled values.

Parameter	Method 2	Method 3
Flutter speed difference [%]	4.16	2.31
Flutter frequency difference [%]	14.52	6.25

Table 5

Relative errors (in %) of Methods 1, 2 and 3 to the target values.

Parameter	Method 1	Method 2	Method 3
Overall mass	7.47	5.71	2.34
Static deflection		1.24	4.01
1st natural frequency	0.98	1.14	0.98
1st mode shape (translational part)	1.46	1.54	1.12
1st mode shape (rotational part)	2.84	3.12	3.21
2nd natural frequency	0.84	0.86	1.98
2nd mode shape (translational part)	1.29	1.47	2.06
2nd mode shape (rotational part)	3.28	3.50	3.34
3rd natural frequency	5.59	6.93	6.12
3rd mode shape (translational part)	5.98	7.12	6.41
3rd mode shape (rotational part)	6.81	7.21	6.12
4th natural frequency	2.70	3.01	3.12
4th mode shape (translational part)	4.01	4.26	2.14
4th mode shape (rotational part)	3.43	3.86	4.11
5th natural frequency	3.63	4.12	4.09
5th mode shape (translational part)	4.15	4.86	5.12
5th mode shape (rotational part)	5.88	6.13	5.46
6th natural frequency	4.82	5.21	4.55
6th mode shape (translational part)	5.68	6.04	6.48
6th mode shape (rotational part)	6.79	7.54	4.98
7th natural frequency	6.36	7.32	5.94
7th mode shape (translational part)	7.16	7.57	6.12
7th mode shape (rotational part)	7.68	8.33	5.21

Despite the far superior computational time (1.9 and 3.2 times more than Method 1 and 2, respectively) required to obtain a scaled model that represents the nonlinear modal behavior, the global average error is lower than with the previous Methods (1 and 2), namely for the higher frequency mode shapes as observable in Fig. 7. Also improvements were found for the mass target parameter. The better matching between mode shapes (full and scaled) is also noticeable from Fig. 7 (b) and (c), which means that Method 3 mimics more suitably the modal behavior of the full scale model. As a consequence of this improvement in the modal behavior, the aeroelastic behavior is also expected to be more properly replicated. The flutter mechanism was found to consist in a coupling between the 4th and 7th mode shape in the full size model, which correspond to the second flapwise bending and first torsion vibration modes of the wing, respectively. Since the vibration modes that comprise the flutter mechanism are more

suitably replicated with the scaled model designed with Method 3, as one can observed in Fig. 7, especially for the 7th mode, the flutter speed and frequency are expected to become closer to the target scaled values, which can be confirmed in Table 4.

Fig. 9(a) and Fig. 9(b) show, respectively, the Fr-g (Froude number vs damping factor of an aeroelastic mode) and Fr-k (Froude number vs reduced frequency) diagrams for the vibration modes that comprise the flutter mechanism, mode 4 (black lines) and 7 (black dashed lines), when linear and nonlinear approaches are applied in order to match the aeroelastic response of the full model. The aeroelastic response agreement improves considerably when the nonlinear approach is applied and the accuracy is noticeably better than the results obtained using traditionally scaled models.

6. Conclusions

Linear and nonlinear approaches to aeroelastic scaling have been reviewed. Truncation of higher frequency modal degrees of freedom using the linear scaling optimization techniques may lead to designs that do not correctly capture the nonlinear kinematic effects. First, two methodologies that describe the linear approach for aeroelastic scaling were implemented and compared. The second method results in lower computational time required to obtain acceptable results for the scaled models.

A new nonlinear approach using Equivalent Static Loads (ESL) for nonlinear static response matching was used in order to help capture structural nonlinearities, which was attained. The nonlinear static deformation achieved with the scaled model designed using Method 3 become significantly closer to the target nonlinear static deformation than when a linear scaling approach was employed. Also improvements in mass, natural frequencies and modes shapes were observed which rendered in a better matching of the dynamic behavior. As a consequence of these improvements, the aeroelastic response agreement was found to be considerably better when the nonlinear approach is applied and the accuracy is noticeably better than the results obtained using the traditional linear scaling methods.

Conflict of interest statement

The authors declare no conflicts of interest with the publication of this work.

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References

- [1] I.H. Abbott, A.E.V. Doenhoff, *Theory of Wing Sections, Including a Summary of Airfoil Data*, Dover Publications, New York, 1959.
- [2] M.Y. Harmin, J.E. Cooper, Aeroelastic behaviour of a wing including geometric nonlinearities, *Aeronaut. J.* 115 (1174) (2011) 767–777.
- [3] R. Palacios, J. Murua, R. Cook, Structural and aerodynamic models in nonlinear flight dynamics of very flexible aircraft, *AIAA J.* 48 (11) (2010) 2648–2659.
- [4] M.J. Patil, D.H. Hodges, C.E.S. Cesnik, Nonlinear aeroelasticity and flight dynamics of high-altitude long-endurance aircraft, *J. Aircr.* 38 (1) (2001) 88–94.
- [5] F. French, F.E. Eastep, Aeroelastic model design using parameter identification, *J. Aircr.* 33 (1) (1996) 198–202.
- [6] J. Katz, A. Plotkin, *Low-speed Aerodynamics*, 2nd edition, Cambridge University Press, Cambridge, 2001.
- [7] R. Bisplinghoff, H. Ashley, R. Halfman, *Aeroelasticity*, Dover Publications, New York, 1996.
- [8] P.A. Ricciardi, *Geometrically Nonlinear Aeroelastic Scaling*, Ph.D. thesis, Faculty of the Virginia Polytechnic and State University, Blacksburg, Virginia, USA, 2013.
- [9] J. Richards, *An Experimental Investigation of a Joined Wing Aircraft Configuration using Flexible, Reduced Scale Flight Test Vehicles*, Ph.D. thesis, University of Victoria, Victoria, BC, Canada, 2014.
- [10] S. Miller, J. Cooper, O. Sensburg, G. Vio, Optimization of a scaled sensorcraft model with passive gust alleviation, in: *Proceedings of 12th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference*, Victoria, British Columbia, Canada, 2008.
- [11] V. Bond, R.A. Canfield, A. Suleman, M. Blair, Aeroelastic scaling of a joined wing for nonlinear geometric stiffness, *AIAA J.* 50 (3) (2012) 513–522.
- [12] Z. Wan, C.E.S. Cesnik, Geometrically nonlinear aeroelastic scaling for very flexible aircraft, *AIAA J.* 52 (10) (2014) 2251–2260.
- [13] A.P. Ricciardi, C.A.G. Eger, R.A. Canfield, M.J. Patil, Nonlinear aeroelastic-scaled-model optimization using equivalent static loads, *J. Aircr.* 51 (6) (2014) 1842–1851.
- [14] J. Richards, A. Suleman, Aeroelastic scaling of unconventional joined wing concept for exploration of gust load response, in: *Proceedings of International Forum on Aeroelasticity and Structural Dynamics, IFASD 2009*, Seattle, USA, 2009.
- [15] E. Buckingham, On physically similar systems: illustrations of the use of dimensional equations, *Phys. Rev.* 4 (4) (1914) 345–376.
- [16] C.L. Pittit, R.A. Canfield, R.L. Ghanem, Stochastic analysis of a aeroelastic system, in: *Proc. 15th ASCE Engineering Mechanics Conference*, New York, NY, USA, 2002.
- [17] P. Pereira, L. Almeida, A. Suleman, V. Bond, R.A. Canfield, M. Blair, Aeroelastic scaling and optimization of a joined wing aircraft concept, in: *Proc. 48th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, Honolulu, Hawaii, USA, 2007.
- [18] G.-J. Park, Technical overview of the equivalent static loads method for nonlinear static response structural optimization, *Struct. Multidiscip. Optim.* 43 (2011) 319–337.
- [19] A. De Gaspari, S. Ricci, A. Antunes, F. Odaguil, G. Lima, Application of active camber morphing concept to a regional aircraft, in: *Proc. 22nd AIAA/ASME/AHS Adaptive Structures, AIAA Science and Technology Forum and Exposition, SciTech 2014*, National Harbor, Maryland, USA, 2014.
- [20] A. Suleman, J. Vale, F. Afonso, F.P. Lau, S. Ricci, A.D. Gaspari, L. Riccobene, L. Cavagna, J. Cooper, L. Lambert, C. Wales, R. Cheung, R. Nangia, A. Rizzi, M. Zhang, H.-P. Monner, B.V. de Kamp, S. Vasista, M. Morelli, J. Parizi, F. Odaguil, G. Lima, A. Antunes, Novel air vehicle configurations: from fluttering wings to morphing flight, in: *11th World Congress on Computational Mechanics*, Barcelona, Spain, 2014.
- [21] O. Galeán, *Optimization and Redesign of the Reaming Beam for Airbus A350 Pylons*, Ph.D. thesis, Universidad Pontificia ICAI ICADE COMILLAS, Madrid, Spain, 2012.